

Work related burnout syndrome, information satisfaction, and concern of safety among Greek nurses during the first wave of COVID-19 outbreak



Original Article

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Received: 3 April 2021; Accepted: 31 August 2021; Published: 20 September 2022

Abstract: **Objective:** To determine the levels of burnout among Greek nurses during the first wave of Coronavirus disease 2019 (COVID-19) pandemic and to investigate the relationship among levels of burnout, satisfaction and concern about information, personal protection equipment (PPE), and safety for COVID-19.

Methods: A cross-sectional study was conducted. Data were collected by using the Burnout subscale of the Greek version of the Professional Quality of Life Scale. Questions concerning satisfaction and concern about information, PPE, and safety for COVID-19 were added. Convenience sampling method was employed. Descriptive statistics and multiple linear regression analysis were applied, and distributions were tested for their symmetry using Bloom's method.

Results: The sample comprised of 301 nurses working in public and private hospitals in Greece. A moderate level of burnout was found among the Greek nurses (mean = 29.95, ± 4.75). Regarding satisfaction and concern about information, PPE, and safety for COVID-19, nurses expressed overall low satisfaction with a mean score of 1.96 (± 0.77), and high levels of concern with a mean score of 3.19 (± 0.85). Increased satisfaction regarding the information provided was significantly associated with decreased levels of burnout ($\beta = -1.32$, $P < 0.001$), and increased concern for COVID-19 was significantly associated with increased levels of burnout ($\beta = 0.83$, $P = 0.010$).

Conclusions: Greek nurses expressed low satisfaction concerning information, PPE, and safety for COVID-19 and high levels of concern of possible COVID-19 transmission. Nevertheless, a moderate level of burnout was found.

Keywords: Burnout syndrome • COVID-19 • nurses • satisfaction • quality

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1. Introduction

At the end of 2019, humanity along with the scientific community faced a Coronavirus disease 2019 (COVID-19) outbreak, caused by a novel virus, SARS-CoV-2. The SARS-CoV-2 causes severe respiratory

How to cite this article: Rovithis M, Kalaitzaki A, Koukoulis S, et al. Work related burnout syndrome, information satisfaction, and concern of safety among Greek nurses during the first wave of COVID-19 outbreak. *Front Nurs.* 2022;3 1–9.

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Argyroula Kalaitzaki

syndrome leading to life-threatening conditions that might even result in death. The outbreak of the virus began in Wuhan, China, and rapidly spread all around the world, a situation that was a particularly aggravating factor for health systems worldwide.¹ By November 2021, >251 million people worldwide have been detected with COVID-19, and >5,077,907 have died.² More than 15% of patients diagnosed with COVID-19 developed severe complications of the disease, with 5%–10% needing to be admitted in an intensive care unit (ICU) due to the severity of the symptoms and the high mortality risk (3%–7%).^{3,4} The pressure for patients' hospitalization in many health care settings has increased excessively. The high transmission rates, the lack of immunity in the community, the delay in the implementation of the tests, the lack of medical material and technical infrastructure, and people's feelings of uncertainty and stress have led an increased number of individuals to seek health services. The health care systems worldwide faced increased and continuous pressure to deal with the COVID-19 cases, and health professionals, particularly the frontline ones, suffered a great deal of physical and psychological burden.^{5–8} In addition, the lack of adequate personal protection equipment (PPE) and staff shortages, as well as shortages of beds and mechanical ventilators, have resulted in increased stress and depersonalization, with detrimental effects on the physical and psychological state of the nurses and other health professionals.^{9–11} Recently, several studies have reported severe impacts of the pandemic on the mental health of the health professionals, with increased depression and anxiety rates because of their inexperience in managing COVID-19 patients.^{12,13} Extremely high rates of burnout and compassion fatigue have been reported among the health professionals who treat COVID-19 patients, and particularly among those who are at the frontline and work in settings with high admission rates of COVID-19 patients, whose outcome is poor.^{14,15}

Exploring the health care providers' professional and emotional exhaustion in various health care settings was an area of research worldwide.¹⁶ Studies on nurses' burnout and emotional exhaustion were also conducted in several countries.^{17–20} In a systematic review and meta-analysis conducted to examine nurses' burnout and associated risk factors during the COVID-19 pandemic, findings showed that the overall prevalence of emotional exhaustion was 34.1%, of which depersonalization was 12.6% and of lack of personal accomplishment was 15.2%. It was highlighted that the main risk factors that increased nurses' burnout were the young age, the decreased social support, the low family and colleagues' readiness to cope with COVID-19 outbreak, and the increased perceived threat of COVID-19, as

well as the high-risk environment involving long working hours, increased workload, and inadequate or insufficient material and human resources.²¹

Studies that have been conducted in Greece during the pandemic era were mainly focused on exploring emotional exhaustion, post-traumatic stress, and associated risk factors among health care professionals.^{22,23} Ilias et al.²⁴ studied the impact of the SARS-CoV-2 pandemic on the mental health of health care professionals in 2 referral hospitals in Athens, Greece. The findings showed that most of the frontline Greek health care professionals appeared to be quite resilient to this challenge, while one-third of them suffered from emotional exhaustion and post-traumatic stress disorder.²⁴ Pappa et al.²⁵ examined the levels and associated risk factors of anxiety, depression, traumatic stress, and burnout of frontline staff in Greece during COVID-19 outbreak. The authors concluded that COVID-19 pandemic adversely affects the psychological well-being of health care workers, demonstrating high prevalence rates of depression, anxiety, traumatic stress, and burnout among Greek frontline staff.²⁵ A cross-sectional study that examined the prevalence of burnout syndrome among intensivists during COVID-19 outbreak in 85 countries, including Greece, highlighted the overwhelming impact of COVID-19 pandemic on ICU specialists, with about half the ICU physicians experiencing symptoms of severe burnout and anxiety.¹⁶

Although Greek nurses are among those health professionals who experience high levels of burnout during the COVID-19 pandemic, there is limited research evidence on this area.²¹ Therefore, the need to investigate the work related burnout syndrome among Greek nurses during the COVID-19 outbreak is imperative to fill in the knowledge gap in this area.

The main aim of the present study was to investigate the levels of burnout among Greek nurses during the first wave of COVID-19 pandemic. Furthermore, the researchers aimed to examine the relationship among levels of burnout, satisfaction and concern about information, PPE, and safety for COVID-19.

2. Methods

2.1. Study design and setting

A cross-sectional study was conducted via a self-reported online questionnaire during May 2020. The study population consisted of registered and auxiliary nurses employed in public and private hospitals in Greece. Inclusion criteria involved at least 3 months of working experience in the hospital. A convenience sampling strategy was employed. The study was advertised to the potential participants via social networks that were

Argyroula Kalaitzaki



addressed to nurses' associations or related to their professional bodies. The online questionnaire was distributed by the Laboratory of Interdisciplinary Approaches to the Enhancement of Quality of Life (Quality of Life Lab), which is affiliated to the Hellenic Mediterranean University. For recruitment purposes, along with the online questionnaire, a detailed information letter explaining the nature, aim, and significance of the study was provided. An informed consent form was also available to all potential participants. The voluntary nature of the study, confidentiality issues, personal data protection, anonymity, and the participants' right to withdraw at any time from the study without any consequences were explained in the online informed consent form prior to data collection phase. Ethical approval was obtained from the Research and Ethics Committee of the Hellenic Mediterranean University. The study sample comprised of 301 nurses (registered and auxiliary), working in public and private hospitals in Greece. None of the participants made the request to withdraw participation at any stage of the research study.

2.2. Instruments

Data were collected with the Greek version of the Burnout subscale of the Professional Quality of Life Scale (ProQoL) (items: 1, 4, 8, 10, 15, 17, 19, 21, 26, and 29). The ProQoL includes 30 items, allocated in 3 subscales of 10 items each, measuring Compassion Satisfaction, Burnout, and Secondary Traumatic Stress.²⁶ The items are rated in a 5-point Likert scale, ranging from 0 (never) to 5 (very often).²⁶ Cronbach's alpha was 0.749.

In addition, 9 questions were formulated by the researchers, out of which 4 were to investigate the degree of satisfaction, concerning: (a) the COVID-19 information provided at workplace, (b) the PPE, (c) the information implementing the COVID-19 protocols, and (d) self-efficacy of treating COVID-19 confirmed cases. Further, 5 questions examined the concern of a possible transmission of COVID-19. Responses were rated in a 5-point Likert scale ranging from 1 "not at all" to 5 "very much" and were summed to produce a total score satisfaction and self-efficacy. Cronbach's alpha was 0.840 for the satisfaction questions and 0.836 for the concern questions. Finally, 9 questions were added to collect the demographic characteristics of the study participants.

2.3. Statistical analysis

Descriptive statistics were presented as means and standard deviations (SDs) for continuous data or frequencies and percentages for categorical data. Descriptive statistics, either means and SDs or frequencies and percentages, were used to present the sample's

characteristics and analyze the data. Data were analyzed using SPSS software (IBM SPSS Statistics for Windows, Version 25.0.; IBM Corp, Armonk, NY, USA). The symmetry test of the parameters under study (Burnout subscale of the ProQoL scales and scores of satisfaction and concern) was performed using the Bloom method (Q-Q plot). Among the questions of the satisfaction and the concern, repeated measures analysis (GLM) was applied. Finally, considering the characteristics of satisfaction of information, protection equipment and other measures of safety required for remaining protected from COVID-19 transmission, and concern about spread of COVID-19 infection, an analysis of the hierarchical multiple linear regression and correlation of the Burnout subscale was performed, in which the unstandardized coefficients of b (unstandardized betas) were evaluated. Significant levels of P values were defined at a critical value of 5%.

3. Results

3.1. Demographic characteristics of the participants

Three hundred and one nurses participated in the study, of whom 80.7% were female, with a mean age of (40.8 ± 9.3) years. Over half of the participants were married and had children. The majority of them was registered nurses (86.4%). They had on average up to 5 years of working experience (26.6%) or 16–20 years (18.5%) (Table 1).

3.2. Burnout syndrome among Greek nurses during COVID-19 outbreak

Analysis showed a moderate level of burnout (mean score 29.95 ± 4.75 ; min 18.00, max 42.00) (results not shown in tables/figures). More specifically, low scores were reported by 6.6% of the participants, average scores by 93%, and high scores by 0.4%.

Regarding satisfaction and concern about information, PPE, and safety for COVID-19, nurses expressed on average low satisfaction, with a mean score of 1.96 ± 0.77 . Overall, high levels of concern (mean score 3.19 ± 0.85) were reported with regard to information on PPE and safety. Nurses appeared to be moderately satisfied with the information received on the procedures to be followed in case of addressing a COVID-19 patient and slightly satisfied with the PPE provided by the hospital (mean scores 2.13 vs. 1.57, $P < 0.01$, respectively). Furthermore, nurses appeared to be very concerned about a possible transmission of the virus to their family and friends, while they reported low concern of being self-isolated due to a possible infection (mean scores 3.81 vs. 2.30, $P < 0.01$, respectively) (Tables 2 and 3).

Items	n	%
Gender		
Male	58	19.3
Female	243	80.7
Age, years, (40.8 ± 9.3, SD = 20.62)		
<40	130	43.2
40+	171	56.8
Marital status		
Married or living with a partner	172	57.1
Single/divorced	129	42.9
Children		
Yes	184	61.1
Level of education		
Diploma	49	16.3
Bachelor degree	135	44.8
Postgraduate	117	38.9
Working experience (years)		
0-5	80	26.6
6-10	31	10.3
11-15	43	14.3
16-20	55	18.3
21-25	35	11.6
26-30	43	14.3
30+	14	4.7
Type of hospital		
Public	171	56.9
Public specialized hospital	22	7.3
University hospital	60	19.9
Private	48	15.9
Departments		
Blood bank	5	1.7
Radiology	3	1
Anesthesia	17	5.6
Laboratories	4	1.3
Wards	95	31.6
Critical care unit	47	15.6
Pediatric intensive care unit	5	1.7
Kidney dialysis unit	10	3.3
Out-patient clinic	3	1
Emergency department	40	13.3
Operation theater	20	6.6
Other	52	17.3
Professional position		
Auxiliary nurse	41	13.6
Registered nurse	230	76.4
Nurse manager	25	8.3
Director	5	1.7

Note: SD, standard deviation.

Table 1. Descriptive characteristics of the study sample.

Table 4 presents the univariate correlations between overall Satisfaction and Concern scores with participants' characteristics. Participants with marital status as married/living with a partner seem to have significantly higher levels of degree of concern about COVID-19 ($r = -0.148$, $P = 0.010$). The higher status of professional position (from Auxiliary Nurse to Nurse Director) is also related with higher satisfaction ($r = 0.114$, $P = 0.049$) and concern about transmission of COVID-19 ($r = 0.123$, $P = 0.033$).

The results of the multiple linear regression with the Burnout scale as an outcome and with the characteristics of the health professionals, information satisfaction, and concern for COVID-19 are presented in Table 5. In the first model, none of the prognostic factors was significantly related to Burnout ($P > 0.05$) while in the second model, increased satisfaction regarding the information provided was significantly associated with decreased levels of burnout ($\beta = -1.32$, $P < 0.001$) and increased concern for COVID-19 was significantly associated with increased levels of burnout ($\beta = 0.83$, $P = 0.010$).

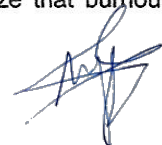
4. Discussion

The present study aimed to investigate the work related burnout syndrome among Greek nurses during the COVID-19 outbreak. Furthermore, the degree of satisfaction and concern regarding information, PPE, and safety for COVID-19 was assessed.

The study results revealed that Greek nurses demonstrate a moderate level of work related burnout during the COVID-19 pandemic. Results from similar studies highlighted that in health crisis situations, the nurses' adherence to their professional role and responsibility tends to prevail over emotional exhaustion, without, however, diminishing the possibility of future burnout incidences.²⁷⁻²⁹ Livadiotis et al.,²⁹ who studied the nurses' psychosocial burden in COVID-19 units in Cyprus, emphasized that during the first wave of the pandemic, nursing staff experienced high levels of work related stress. However, burnout symptoms were not reported, and nurses appeared to be highly attached to their professional role.

Wu et al.³⁰ compared burnout syndrome frequency among physicians and nurses working in the frontline and in usual wards during the pandemic outbreak in Wuhan. Results indicated that the frontline health professionals had a significantly lower frequency of burnout during the COVID-19 crisis than those working in usual wards. The authors further state that frontline health professionals who directly address COVID-19 may have a greater sense of control in their workplace, which is explained as a major effort of engagement as well as an important factor for avoiding burnout. On the contrary, Hoseinabadi et al.³¹ emphasize that burnout

Argyroula Kalaitzaki



Field and questions	Not at all satisfied (1)	Slightly satisfied (2)	Moderately satisfied (3)	Very satisfied (4)	Completely satisfied (5)	Mean Score	SD
Satisfaction with							
1. The COVID-19 information provided at workplace	31.2	36.5	22.3	9	1	2.12*	0.99
2. The information of implementing the COVID-19 protocols	32.6	35.9	20.3	10	1.3	2.13	1.02
3. PPE	60.5	25.6	10.6	3	0.3	1.57	0.82
4. Overall self-efficacy of treating COVID-19 confirmed cases	32.6	38.5	23.6	4.7	0.7	2.02	0.9
Total Satisfaction Score						1.96	0.77

Note: COVID-19, Coronavirus disease 2019; PPE, personal protection equipment; SD, standard deviation.

*GLM repeated measures analysis among the questions of the satisfaction and the concern: $P < 0.001$.

Table 2. Satisfaction regarding information, PPE, and safety during COVID-19 pandemic.

Concern	Never (1)	Rarely (2)	Sometimes (3)	Often (4)	Always (5)	Mean Score	SD
I am afraid of							
1. Being infected from COVID-19	5.6	32.6	37.9	14	10	2.90*	1.04
2. Transmitting the virus to my family/friends	1.7	7.3	36.2	18.3	36.5	3.81	1.07
3. Being self-isolated due to infection	28.6	32.9	24.6	8.3	5.6	2.30	1.34
4. Transmitting the virus to my colleagues	4.7	17.3	37.2	19.9	20.9	3.35	1.13
5. Transmitting the virus to my patients	4.3	8.3	38.9	20.3	28.2	3.60	1.11
Total Concern Score						3.19	0.85

Note: COVID-19, Coronavirus disease 2019; PPE, personal protection equipment; SD, standard deviation.

*GLM repeated measures analysis among the questions of the satisfaction and the concern: $P < 0.001$.

Table 3. Concern regarding information, PPE, and safety during COVID-19 pandemic.

Items	Score of satisfaction, <i>r</i> -Pearson (<i>P</i> value)	Concern
Gender (1: men, 2: women)	-0.060 (0.300)	-0.090 (0.118)
Age (years)	-0.008 (0.886)	-0.022 (0.707)
Marital status (1: married/living with a partner, 2: single/divorced)	0.059 (0.310)	-0.148 (0.010)
Children (number)	0.073 (0.206)	-0.103 (0.826)
Level of education (1: diploma, 2: bachelor degree, 3: postgraduate studies)	0.137 (0.017)	0.037 (0.527)
Working experience (years)	0.012 (0.837)	0.030 (0.606)
Professional position (1: auxiliary nurse, 2: registered nurse, 3: nurse manager, 4: nurse director)	0.114 (0.049)	0.123 (0.033)

Table 4. Pearson correlation between overall Satisfaction and Concern scores participants' characteristics.

levels appeared to be higher for frontline nurses compared to those who work in usual wards. Nevertheless, the authors state that work related stress significantly influences the burnout levels, while the sense of successfully controlling ongoing and future pandemic outbreaks may be a factor in preventing nurses' burnout. In addition to that, Manzano and Ayala²⁰ demonstrate

that organizational support, such as clear, fluid, and regular communication with the nursing staff, would help increase nurses' confidence and sense of control and thus prevent work related burnout.

The participants in the present study showed moderate levels of satisfaction with the information received regarding the procedures that should be followed in the

Argyroula Kalaitzaki



Prognostic factors	Burnout Scale ^a			
	First model		Second model	
	β (95% CIs)	P value	β (95% CIs)	P value
Gender (1: men, 2: women)	0.54 (-0.84, 1.91)	0.442	0.59 (-0.74, 1.93)	0.384
Age	0.06 (-0.05, 0.16)	0.296	0.07 (-0.04, 0.17)	0.203
Marital status (1: married/living with a partner, 2: single/divorced)	-0.89 (-2.02, 0.24)	0.13	-0.51 (-1.62, 0.59)	0.364
Level of education (1: diploma, 2: university, 3: postgraduate studies)	-0.01 (-0.88, 0.86)	0.982	0.08 (-0.46, 0.63)	0.846
Working experience (years)	-0.04 (-0.54, 0.47)	0.884	-0.08 (-0.57, 0.41)	0.748
Professional position (1: auxiliary nurse, 2: registered nurse, 3: nurse manager, 4: nurse director)	-0.42 (-1.46, 0.62)	0.429	-0.20 (-1.21, 0.81)	0.702
Total Information Satisfaction Score – COVID-19 ^b			-1.32 (-2.01, -0.63)	<0.001
Total Concern Score – COVID-19 ^c			0.83 (0.20, 1.46)	0.01
R ² adjusted	0.003		0.072	

Note: CIs, confidence intervals; COVID-19, Coronavirus disease 2019.

^aScale from 1 = never to 5 = very often, where the mean of the subscale from the respective questions emerged, with a high score indicating a greater degree of exhaustion.

^bRating of answers from 4 questions: 1 = not at all to 5 = absolutely (average score of 4 questions).

^cRating of answers from 5 questions: 1 = not at all to 5 = too much (average score of 5 questions).

Table 5. Hierarchical multiple linear regression of the Burnout Scale as an outcome with the characteristics of Information Satisfaction and Concern for COVID-19 in the study participants as independent variables.

case of addressing a possible COVID-19 patient and a low degree of satisfaction with the PPE provided by the hospital. Livadiotis et al.²⁹ highlighted that although nurses' satisfaction with the provided information was moderate, satisfaction with PPE was high. This discrepancy may be due to the degree of readiness of each hospital or country to provide the necessary PPE and support to health professionals. Furthermore, Mo et al.,³² who investigated the work related stress among Chinese nurses during the pandemic outbreak in Wuhan, emphasized that safe working conditions for nurses and provision of information may contribute significantly toward preserving the mental health of frontline nurses. Similar results are reported by Wu et al.,³⁰ who demonstrated the positive impact of information and training on nurses' mental health. Finally, Said and Shafei³³ stressed that providing timely and clear information on the ideal means to manage COVID-19 patients and availability of resources help nurses attenuate their personal anxieties and fears.

Regarding the nurses' levels of anxiety as recorded in the present study, the participants expressed a high degree of fear pertaining to possible transmission to their family and friends and less concern regarding the possibility of personal infection and self-isolation. Similar results were demonstrated in a qualitative study,³⁴ which explored nurses' experiences of psychological distress during the care of patients with COVID-19. Results of this study revealed that nurses experienced anxiety due to the nature of the disease, death anxiety, and fear of being

contaminated and of infecting their families.³⁴ Another relevant study³⁵ examined the frontline nurses' burnout, anxiety, depression, and levels of fear in addition to their associated factors during the COVID-19 outbreak in Wuhan, China. Findings revealed that nurses experienced a moderate level of burnout and a high level of fear.³⁵ The number of colleagues who died from COVID-19 or related causes, such as fatigue and overwork, appear to have aggravated their fears. These issues are associated with nurses' fear of infection and death as well as the fear of spreading the virus to their loved ones.

The results of the present study were associated with the marital status of married/living with a partner and also with the higher status of professional position (from Auxiliary Nurse to Nurse Director). More specifically, nurses who were married or living with a partner demonstrated higher levels of concern compared to those who were single or divorced ($P = 0.010$). Additionally, both the degree of satisfaction and concern appeared to be higher for nurses who were university graduates and/or nurse managers. This is probably due to the fact that the professional positions of high responsibility (managers, supervisors, directors), create a stressful environment of applying highly demanding management interventions, as well as immediate and high-risk decision-making processes. Similar results were presented in the study of Hu et al.,³⁵ in which the levels of burnout, anxiety, depression, and fear were positively correlated with nurses' professional position. On the contrary, Livadiotis et al.²⁹ assert that frontline

Argyroula Kalaitzaki



nurses experience higher levels of burnout compared to nurse managers, while in the study of Labrague and de los Santos,³⁶ nurses who had part-time job roles experienced higher levels of fear of COVID-19 infection.

Furthermore, the results of the present study indicate that the degree of satisfaction and concern with regard to information is closely associated with burnout syndrome. Increased satisfaction with the information received for COVID-19 was significantly associated with reduced mental exhaustion and burnout, while increased concern about COVID-19 appeared to be significantly associated with increased levels of burnout. The importance of adequate information and training during the COVID-19 outbreak was mentioned in previous studies, which demonstrated that safe working conditions for nurses and provision of information and training may mobilize nurses' conscientiousness and support their mental health.^{30,32} Similar results were reported by Arnetz et al.,³⁷ who demonstrated that organizational interventions such as education and training are important for reducing levels of concern and anxiety and maintaining nurses' health, safety, and well-being. Specifically, the failure of the Health Care Organization to fulfil the nurses' educational needs regarding caring for COVID-19 patients was reported as a factor that increased nurses' stress and anxiety. Further research showed that the type of information provided during the pandemic may affect nurses' psychology. For instance, receiving negative or neutral information about the pandemic was referred to as a major factor for a higher risk of developing anxiety, insomnia, and related psychological problems.³⁸ Finally, Heath et al.³⁹ summarized that competency training, education, and training focused on resilience, as well as staff feedback sessions, are key interventions for minimizing the risk of psychological distress and burnout for health care professionals working with COVID-19 patients.

Specific limitations can be mentioned in relation to this study. The convenience sample and the underrepresentation of certain subgroups (e.g., males) limit the generalizability of the findings. The web-based survey may have contributed to selection bias. Social

desirability bias could have been another limitation. The results should be considered in light of the fact that study participants were comprised of both registered and auxiliary nurses, 2 groups with diverse responsibilities concerning the provision of patient care. Comparisons between frontline and non-frontline nurses could have clarified further the factors contributing to increased risk of burnout. Despite the limitations, the study demonstrated useful insights about the psychological impact of the COVID-19 outbreak on Greek nurses' mental health and burnout, assessed and tackled in similar situations.

5. Conclusions

During the first wave of COVID-19 outbreak, Greek nurses experienced a moderate level of burnout, low satisfaction concerning information, PPE, and safety and high levels of concern of possible COVID-19 transmission. Additionally, nurse managers and health professionals in higher professional posts experienced increased levels of stress and anxiety. Targeted organizational interventions such as adequate staffing, provision of appropriate PPE, safety, information, and training are important for supporting health care professionals who work with COVID-19 patients. Early detection of health professionals' psychological distress and supportive interventions are essential to successfully address the growing health care needs and to prevent adverse psychological effects on frontline nurses during COVID-19 pandemic.

Ethical approval

Ethical approval was obtained from the Research and Ethics Committee of the Hellenic Mediterranean University (Ref No. 13/07-04-2020).

Conflicts of interest

All contributing authors declare that no conflicts of interest exist.

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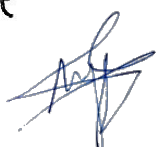
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Argyroula Kalaitzaki



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